

"However, both types of modules have a higher range and better penetration than Wi-Fi. Coupled with an efficient mesh protocol, RF modules find applications in all smartcity domains like smart metering and smart lighting."

He adds, "In the Asian scenario in general and the Indian scenario in particular, solutions should use a combination of technologies. As mentioned earlier, concrete obstacles are aplenty and Internet connectivity is patchy in some areas. This calls for a solution that has an RF mesh to collect data from devices like utility meters, lighting fixtures and home sensors. This mesh would then transmit data to a gateway device. Using an appropriate connection backbone like Wi-Fi, Ethernet, WiMAX or GPRS, the gateway will then upload data to the Internet. The gateway device and RF modules need to be able to work even in the absence of Internet connectivity to ensure that the solution is not Internet-dependent."

Praveen Ganapathy, director - embedded processing, India sales and applications, Texas Instruments India, adds, "We see broadest traction for BLE across segments, followed by Wi-Fi for home and enterprise segments, followed by sub-GHz radios with support for Zigbee, 6LoWPAN and more in the industrial space. Near Field Communication is also seeing increased traction in payments, pairing and related applications involving authentication."

Anand informs, "All Wi-Fi and RF modules are IoT-centered these days. However, not all fit due to the fact that some criteria do not match. For example, both NodeMCU and smartWi-Fi module are based on the same chipset, however, smartWi-Fi has a battery charger and a user button, LED on board, while NodeMCU does not, that makes the final setup much bulkier."

Advantages of using Wi-Fi modules in IoT

There are many wireless interface options like Bluetooth Low Energy, ZigBee, Z-Wave, Wi-Fi and RFID, each with its own unique balance of power, range, data rates, mesh networking, interference immunity and ease of use. However, some interfaces are not yet native IP enabled, so cannot be addressed directly or exchange data with other devices and servers over the Internet. These require a separate gateway, adding expense and complexity to the final solution.

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Wi-Fi is based on IEEE 802.11 standard with native IP addressability, is ubiquitous, well understood and can scale well in terms of data rates to optimise for power consumption. 802.11 standards are also IPv6 compliant, so there is almost no limit to the number of unique addresses.

Microchip supplies an IoT development kit (DM990001) using a module driven by a 32-bit controller. It employs two Microchip components, MRF24WG0MA/B pre-certified Wi-Fi module, which support both 802.11b and 802.11g, and PIC32MX695F512H MCU.

The starter kit powered by Amazon Web Services (AWS) is a managed Cloud platform that lets connected devices securely interact with Cloud applications. With AWS, IoT applications can keep track of and communicate with all networked devices, all the time, even when these are not connected.

Texas Instruments CC3100MOD, an FCC, IC, CE and Wi-Fi certified module, is a part of the new SimpleLink Wi-Fi family that dramati-

cally simplifies implementation of Internet connectivity. It integrates all protocols for Wi-Fi and the Internet, which greatly minimises host MCU software requirements.

STMicroelectronics Intelligent Wi-Fi modules represent plug-and-play, standalone 802.11 b/g/n solutions for easy integration of wireless Internet connectivity features into existing or new products.

The future

Like most popular wireless technologies, Wi-Fi is constantly in a state of development. The latest iteration being rolled out is called 802.11ac, which provides rates up to 1.3Gbps in the 5GHz unlicensed band. Most access points, home routers and smartphones do not have it yet, but it is working its way into all of these.

Also underway is the process of finding applications other than video and docking stations for the ultrafast 60GHz (57GHz-64GHz) 802.11ad standard. It is a proven and cost-effective technology, but do we need 3Gbps to 7Gbps rates up to ten metres?

Wireless is certainly the future for the IoT and machine-to-machine communication. Wi-Fi is an obvious choice because it is ubiquitous, but it is overkill for some apps and a bit too power-hungry for others.

Bluetooth is another good option, especially Bluetooth Low Energy. Bluetooth's new mesh and gateway additions make it even more attractive.

ZigBee is a ready-and-waiting alternative; so is Z-Wave. Then there are multiple 802.15.4 variants like 6LoWPAN.

As Wi-Fi chips get smaller with more features and solve the connection process problem, we can expect to see more IoT devices in the wild, monitoring the air, our health and more, as well as controlling physical objects from the Internet. **EFY**